Noname manuscript No.
(will be inserted by the editor)

MFGAN: Towards a Generic Multi-Kernel Filter based Adversarial Generator for Image Restoration

Abderrazak Chahi* · Mohamed Kas* · Ibrahim Kajo* · Yassine Ruichek

Received: date / Accepted: date

Abstract In recent years, there has been a growing interest in the use of Generative Adversarial Networks (GANs). Thanks to their outstanding performance in image translation and generation, they play an increasingly important role in computer vision applications. Most approaches based on GAN focus on proposing task-specific auxiliary modules or loss functions that are tailored to address various challenges of a single application, but often do not perform better when evaluated to improve other image generation tasks. Moreover, the basic ResNet and U-Net based GAN generators reach their limits in many image restoration and enhancement use cases. Therefore, in this paper, we propose a generic GAN referred to as Multi-Kernel Filter-based Conditional Generative Adversarial Network (MFGAN). We develop a new GAN generator with multiple CNN streams to extract more relevant and discriminative features related to the studied task. The proposed MFNet generator consists of two CNN modules, feature extraction and feature compression, which are combined to connect both the GAN encoder and decoder. It considers the strengths of conventional layers at different scale levels with multi-kernel filtering to capture high to low feature frequencies that reflect the complex image degradations and structural image details. Extensive experiments on five challenging applications for image enhancement, image restoration, and infrared image translation demonstrate the superiority and effectiveness of the proposed MFGAN in removing image degradation and generating visually appealing fake images. Our MFGAN quantitatively outperforms both state-of-the-art GANs and other CNN-based architectures in all tested benchmarks.

Keywords Generative adversarial network · Image translation · Image restoration · Generic adversarial generator · CNN branches · Supervised image generation · Feature extraction · Feature compression · Multi-kernel filtering

Abderrazak Chahi, Mohamed Kas, Ibrahim Kajo, Yassine Ruichek.
CIAD, UMR 7533, Univ. Bourgogne Franche Comté, UTBM, F-25200, Montbéliard, France.
E-mail: abderrazak.chahi@utbm.fr, E-mail: mohamed.kas@utbm.fr, E-mail: ibrahim.kajo@utbm.fr, E-mail: yassine.ruichek@utbm.fr.

*Authors contributed equally.

1 Introduction

Generative Adversarial Networks (GANs) have received mo-re attention in the deep learning research community since their introduction in [1]. Their ability to generate and manipulate data in a variety of domains has contributed greatly to their success. Due to their outstanding performance in image translation and generation tasks, they are playing an increasingly important role in computer vision applications. GANs are designed as a theory sense game between two neural networks trained in an adversarial manner to achieve a zero-sum Nash equilibrium profile [2]. In other words: When one network wins, the other loses, as in a minimax game. The GAN model converges when both networks reach an optimal point where one network no longer updates its weights, regardless of what the other network does. The first network, the discriminator (denoted D), is a binary classifier that learns to distinguish, as accurately as possible real-world data from instances generated by the second network, the generator (G). In contrast, the generator seeks to fool the discriminator (D) and tries to generate realistic data that cannot be distinguished by the discriminator as

fake instances which in turns also enhances the prediction performance of the discriminator.

Thanks to their efficiency, which is due to their selfsupervised architecture, GANs are used in almost all applications introduced to generate images according to predefined targets. These applications include image-to-image translation [3,4], human pose synthesis [5], image manipulation [6,7], image super-resolution [8], video applications in computer vision and graphics communities [9,10], sketch synthesis [11], facial landmark detection [12], facial aging [13], medical analysis [14], and semantic segmentation [**?**].

Based on the target application, GANs generate images according to two main approaches: 1) unsupervised image generation and 2) supervised image generation. Unsupervised learning addresses the problem of unpaired image-to-image translation where the network attempts to model the mapping from one domain (source) to another (target) with unpaired training data [15,16,17]. The most common architectures that use unpaired adversarial learning are StyleGAN [18] and CycleGAN [19], which are widely referenced in the literature. The second approach is based on the pix2pix architecture [20] that uses paired input and target (ground truth) images from the same domain without pixel annotation, compared to classical CNN frameworks [21,22,23].

For image generation/enhancement tasks, most of the existing GANs frameworks use the classic ResNet [24] or U-Net [25] architectures or a combination of both of them as base generators. Such architectures are easy to build but have performance limitations and produce low quality images in some scenarios. On the one hand, these models can suffer severely in terms of network convergence as they remain unstable, network generation leads to collapse mode as only a limited varieties of instances are generated, or the generator/discriminator is unbalanced leading to network overfitting. Also, during the learning process, the discriminator may become too successful to achieve its goal, causing the generator model to saturate and not learn anything as the gradient regularization vanishes.

On the other hand, there have been several attempts to improve GANs where researchers focused on improving the loss function without changing the architectural generator modules that can provide and extract more discriminant and conditional features if properly designed. The related works that are proposed to improve the GANs loss function, mainly focus on proposing additional penalties to enforce constraints for a better optimization objective. Other GANs-based approaches have only used the pix2pix [20] and CycleGAN [19] architectures with their ResNet and U-Net based standard generator models to develop new datasets that are publicly available for evaluation and training purposes. Considering all these concerns, we believe that developing a well-designed generator architecture will further improve the GANs performance. Therefore, in this paper, a novel generator, referred to as Multi-Kernel Filter Generative Adversarial Network (MFGAN), is proposed to further improve conditional GANs' performance while also motivating the discriminator to improve its regularization and convergence learning speed.

It is important to note that EfficientNet, Vision Transformers, DenseNet, or other models have not, to the best of our knowledge, been extensively explored or used in GAN frameworks, especially as generator models. While these models have shown good performance in image classification tasks, their application and effectiveness specifically as generator models in GANs is still relatively unexplored or incompatible with GAN. Therefore, to maintain consistency with commonly used baseline models and to enable meaningful evaluation and comparison in the context of our study, we decided to compare our proposed MFGAN generator model with the established ResNet and U-Net architectures. By focusing on these baseline models, we aim to provide a solid benchmark and ensure a fair evaluation of the performance and novelty of our proposed generator model.

The proposed generator network consists of four different CNN streams, all of which are stacked to extract perceptual and discriminant features from the input images. In the four CNN branches, different sliding convolutional layers with varied filter sizes are used to limit the sensitivity to network initialization and regulate the overall learning process to mitigate the problems of gradient exploding and collapse mode. A concatenation layer is used to combine the outputs of the four CNN branches, resulting in a holistic deep feature map with detailed image information. Although our approach can be used in both supervised and unsupervised learning manner, the focus here is on supervised image generation/enhancement and translation. To validate and evaluate the superiority of the proposed GAN architecture, we have selected five different challenging applications for image translation and enhancement, and conducted extensive experiments with different benchmarks. The five applications are image demoireing, infrared image translation, dewatering, JPEG artifact removal, and image denoising. Existing CNN models and GANs have already been used for these applications, allowing us to compare our architecture with the current state-of-the-art. Quantitative and qualitative evaluations show that the proposed method is a reliable solution to improve the five selected applications com-

pared to the current SOTA. The main contributions of the present work are highlighted below:

1. We proposed a new generative adversarial network, called Multi-Kernel Filter GAN (MFGAN), for image generation and enhancement.
2. We design a generic and reliable generator network to improve feature extraction for better quality image generation.
3. The proposed MFGAN model is a generic plug-and-play network that supports any autoencoder architecture thanks to its adaptive kernel-filter mapping.
4. Extensive experiments with five challenging benchmarks for image generation and enhancement demonstrate the effectiveness and superiority of MFGAN compared to the current state-of-the-art.

2 Related works

Convolutional Neural Networks (CNNs) and Generative adversarial networks (GANs) have become a natural choice for image generation and enhancement-based applications. In this section, we present and discuss the well-known CNN and GAN -based approaches used to improve these applications. We also discuss the limitations of the traditional Resnet and U-Net-based GAN -generators and explain our motivation for proposing a novel generator architecture.

2.1 Visible-to-infrared translation

The first targeted application deals with the translation of visible images into synthetically generated infrared images. GANs are practical methods to synthetically convert visible images into these types of low-light images. The most challenging dataset used in our experiment is the visible-infrared paired LLVIP dataset presented in [26]. The SOTA method evaluated on the LLVIP dataset is the well-known pix2pix GAN approach. Quantitatively, it performs poorly in terms of PSNR and SSIM, and qualitatively, the background of the generated image and the pedestrian and car shapes are cluttered, the details are wrong and interspersed with artifacts. Recently, Liu et al. [27] expanded upon the pix2pix architecture by incorporating a pyramid structure that operates at the original image resolution. This enhancement enables the model to process multiple scales or resolutions of the input image, allowing it to capture information at various levels of features. One of the significant challenges encountered when translating visible images arises from the fact that infrared images are obtained based on the temperature field of an object's surface. This characteristic poses difficulties in accurately discerning objects in low-light conditions.

2.2 Image demoireing

Image Demoireing is the second application considered in our experimental study. It restores clean versions of images corrupted by multiple moiré patterns. The interference between the pixel grid of the camera and the screen of the electrical device causes this image degradation. The most commonly used benchmark to evaluate SOTA systems for moiré pattern removal is TIP2018 [28], which is considered in our experiments. Convolutional Neural Networks (CNNs) have been used to improve the task of image demoireing. CNN models were first used by Sun *et al.* [28] to automatically remove moiré patterns from images. They proposed a multiresolution end-to-end CNN (DMCNN) to convert an input image into multiple feature maps at different resolution levels by using a stack of cascaded convolutional layers. In this way, the moiré pattern component in each frequency band was removed separately. They provided a large benchmark for training and testing, taken from LCD screens in the ImageNet dataset. The DMCNN method was compared with SOTA methods such as VDSR [29], DnCNN [30], and RTV [31], which were originally used for image denoising, super-resolution, and texture removal. At that time, the DMCNN method performed better than the SOTA methods mentioned above, but showed a significant drop in performance for moiré images with strong large-scale color streaks, such as those taken in the wild. Another limitation reported in [28] is that the DMCNN method could not significantly reduce the blur in the input images due to the degraded components caused by high-frequency moiré patterns and the lack of specific constraints for dealing with such extreme patterns. In a later work in [32], a Dynamic Feature Encoder (DFE) based adaptive instance normalization [33] was introduced to improve the previous DMCNN method. He *et al.*[34] proposed a moiré pattern removal neural network (MopNet) consisting of three core moiré pattern removal components. These include the multiscale feature aggregation to deal with the artefact frequency distribution, the channel-wise target edge predictor to account for the magnitude of moiré effects in the colour channels of the image, and the attribute-based classifiers to characterise the dominant frequency, colour, and shape of the pattern. Note that He *et al.* [34] annotated the dataset with attribute labels of moiré patterns, which helped the network learn complex patterns. However, the MopNet network showed limitations in some scenarios where the output still contains

irregular textures such as gravel or pavement in the background, which affected the overall performance and resulted in incomplete moiré removal. Recently, the work of [35] presented an end-to-end multiscale bandpass convolutional neural network (MBCNN) for image demoireing. The MBCNN network consists of two steps: moiré texture removal using a learnable bandpass filter (LBF) and image color restoration using global/local tone mapping. In this approach, the RGB range is treated in multiple scales. Such a strategy has limited ability to correctly distinguish moiré patterns from the structural details of the image.

2.3 Underwater image enhancement

The third application deals with color tube degradation in underwater images. This includes low contrast, color shift, and like-haze degradation. This is due to poor shooting conditions, such as lighting conditions, type of water, and depth. Several approaches have been proposed to improve underwater images to cope with the absorption and scattering caused by the physical properties of the underwater environment, most of which are based on different variants of GAN. The benchmark selected for this task is the image enhancement dataset (UIEBD) [36]. The Underwater Generative Adversarial Network (UGAN) was proposed by [37] to improve the quality of underwater visual scenes. They used the U-Net network as the base generator and the PatchGAN model as the discriminator, trained on image pairs generated by CycleGAN, and tested on the UIEBD dataset. However, their method still results in contrast degradation and color shifts in some scenes. Li *et al.* [38] proposed a generative adversarial fusion network (FGAN) for image dewatering. Their model consisted of two inputs to effectively correct image color and showed faster test time with fewer parameters compared to the Cycle Consistent Generative Adversarial Network (CycleGAN) evaluated in [39] and the underwater model GAN (UGAN) [37]. Recently, Li and Zhuang [40] presented an adversarial fusion network called DewaterNet. They used multilevel objective functions, including generator loss, fusion-enhanced image loss, SSIM loss, and PSNR loss, to correct color shifts and improve image quality through spectral normalization.

2.4 Image denoising

The fourth application is image denoising to remove unwanted detail from a noisy image. Since noise patterns are high frequency components, it is still difficult to separate them completely from the image, hence the denoised image output may eventually lose some structural details. In our experiments, we selected the well-known Color CBSD68 dataset [41] to evaluate the effectiveness of our method in image noise removal. With the advent of deep learning, Convolutional Neural Network (CNN) has been widely used for image denoising. [42] proposed a deep fully convolutional encoding- decoding framework (RED30) with symmetric skip connections. Convolutional and deconvolutional layers are combined to model the denoising task by extracting the abstraction of the image content while removing noise and recovering structural details of the image. Zhang *et al.* [43] employed an ensemble of CNN denoisers and integrated them into a model-based optimization approach to improve image denoising and reduce processing time. Santhanam *et al.* [44] proposed a recursively branched deconvolutional network (RBDN) that aggregates the multicontext representation of the image using a recursive branching scheme with parameter sharing and adaptive upsampling. One of the drawbacks of the RBDN network is the use of the MSE loss function, which is known to be inappropriate perceptual loss function for tasks that require visually appealing enhanced image output. Liu *et al.* [45] combined CNNs and the wavelet transform originally used for signal processing to improve image denoising. The proposed multi-level wavelet CNN (MWCNN) reconstructs high-resolution feature maps and then cleans up the noisy image by adding CNN blocks between every two levels of the discrete wavelet transform (DWT) via pooling layers. Li *et al.* [46] improved the DenseNet model using wide and transformed network to learn pixel distribution features from noisy images. Compared to traditional methods, their method was successful in removing noise. Recently, an end-to-end attention-driven denoising convolutional neural network (ADNet) was presented in [47], which consists of a sparse block to reduce the depth of the denoising network, a feature enhancement block to capture discriminative global and local features in the degraded image, an attention block (AB) to control the CNN to learn the noise details, and a reconstruction block (RB) to restore the clean image. While increasing the depth of CNN models can improve performance to some extent, it also makes it difficult to train the model efficiently and can lead to collapse mode or the vanishing gradient problem.

2.5 JPEG artifact removal

The last image restoration application we considered in our study is JPEG artifact removal. JPEG com-

pression is widely used thanks to its ability to significantly reduce the file size of images. However, high compression ratios can result in noticeable artifacts that degrade image quality. These artifacts can take the form of blocky or blurry regions, ringing or halo effects at the edges, and loss of fine detail in the image. Removing JPEG artifacts while recovering the coefficients of the image's high-frequency details is a challenging task in image enhancement. This motivates researchers to develop deep learning approaches aimed at removing artifacts and improving the quality of JPEG images to support various applications that rely on visual data. Deep-learning-based approaches are significantly superior to conventional soft decoding methods in terms of both recovery performance and runtime. Here, we provide an overview of current approaches for JPEG artifact removal. Yu et al. [48] exploited the capabilities of CNNs to acquire a mapping between compressed images containing artifacts and their corresponding artifact-free versions. Their findings demonstrated that leveraging shallow features contributes to the convergence of their proposed network, known as ARCNN, resulting in improved removal of JPEG artifacts. Another contribution to image restoration was presented by Yunjin Chen and Thomas Pock [49], who proposed an innovative technique called trainable nonlinear reaction diffusion (TNRD). This approach models the image restoration process as a reaction-diffusion system in which the diffusion process is controlled by a set of trainable parameters, including linear filters and influence functions. By incorporating these parameters and using appropriate reaction forces, this framework enables adaptive and data-driven restoration that effectively addresses various image degradation scenarios such as image denoising and JPEG artifact removal. Deep CNNs integrated with residual learning have demonstrated remarkable advancements in eliminating JPEG artifacts in comparison to handcrafted techniques. The concept of residual learning originates from the understanding that neural networks can often learn the residual, i.e., the disparity between a degraded image and its clean counterpart, more effectively than directly learning the clean image itself. Several variants of deep CNNs employing residual learning have been proposed, such as DnCNN (Denoising Convolutional Neural Network) [30], ARN (Adaptive Residual Networks) [50], and RED (Residual Encoder-Decoder) [51], among others. These networks are typically composed of multiple convolutional layers interconnected with residual connections, which facilitate enhanced gradient flow during training and enable more efficient acquisition of residual information.

2.6 Motivation and discussion

Improving the performance of GANs has attracted much attention since the first GANs were proposed. The U-Net [25] and ResNet block [24] autoencoder architectures are the most commonly used base generators because they are simple and provide good performance for some tasks. The U-net architecture was originally introduced for biomedical image segmentation [25]. Its main concept is to use successive and symmetric contracting layers (encoding) to capture feature representations at different levels, immediately followed by the upsampling operators (decoding) to project the discriminative features learned by the encoder (lower resolution) onto the pixel space (higher resolution), giving the final architecture the shape of a "U" and hence its name. All U-Net variants have the same pipeline, the only difference being the supported image size, which is controlled by the amount of GPU memory, where most frameworks considering 256 and 128 resolutions. Although the U-net is a significant achievement in the field of deep learning, it still suffers from the following limitations: (1) The architecture of the U-net relies on staking out encoding and decoding layers and making skip connections between them, which requires extensive architectural search or an inefficient ensemble of models of different depths. This fact makes the U-net vulnerable to gradient vanishing, especially when the input and target images do not share many features (domain translation); (2) Since U-net skip connections require and enforce aggregation only for the feature maps of the encoder and decoder subnets at the same level, they are an unnecessarily restrictive fusion scheme. Moreover, the late skip connections (last decoding layer) lead to pixel outliers with values exceeding the normalized range and causing artifacts in the output image; (3) Another bottleneck of the U-net is the instability and slowness of the training due to the large number of parameters to be optimized. Also, the overall model suffers from several convergence difficulties due to the lack of feature extraction blocks.

The ResNet-based generator solves the deep feature extraction problem by inserting residual blocks between the encoder and the decoder. However, these residual blocks are redundant and share the same layers and kernel sizes. This topology causes the model to produce blurred images despite its feature extraction sub-blocks. In addition, the residual connection from the first ResNet block to the last block causes the decoder to process the encoded filters itself, which is similar to the U-net problem when the images come from different domains. The kernel size of 3×3 used in the convolutions in the residual blocks with a stride size of 1 results

in vertical patterns in the output image in the form of columns with pixel outliers. Using a small kernel size and a stride size of 1 means that each pixel of the input image is processed twice and therefore the values of its filter maps are duplicated.

As can be concluded from the discussed related works on the five target applications, the majority of the approaches based on GAN focus on proposing task-specific submodules or additional penalties in the cost function to restrict GAN to the improvement of the task under study. In other words, dedicated architectures or loss functions are specific to the different challenges of a particular application but often do not achieve better performance when evaluated to improve other image enhancement tasks, making them a non-generic solution. Moreover, most SOTA work has used only the pix2pix [20] and CycleGAN [19] architectures with their ResNet [24] and U-Net [25] baseline generator models, which reach their limits in many image enhancement tasks. As far as we know, there is no approach that attempts to develop a new and effective generator architecture to improve the overall performance of GAN. Therefore, the development of a GAN generator with dedicated CNN streams that takes into account the strengths of conventional layers at different scale levels would be very useful for a better characterization of features and for a better optimization goal. To overcome the challenges of SOTA, we propose a practical and generic GAN generator called MFNet to improve the performance of GAN in various use cases of image generation and enhancement. The next section is devoted to presenting the proposed methodology.

3 Proposed method

In this paper, we propose a new and reliable generator architecture called MFNet to improve the performance of GAN in various image generation and enhancement applications. This section is divided into subsections to comprehensively present the overall architecture of the proposed framework, the configurations of the MFNet generator and discriminator networks, and the adversarial loss function.

3.1 Generative Adversarial Network

In Goodfellow's original work on generative adversarial networks [1], GAN was introduced as a mini-max game between two different players (i.e., discriminator and generator). The generator attempts to minimize the loss of generated images that are classified as fake by the discriminator, and to optimize the discriminator's performance by punishing it for classifying a fake image as real by maximizing its associated loss. The GAN loss function, commonly known as min-max loss, is expressed as follows:

$$\begin{aligned}\min_G \max_D \mathcal{L}_{cGAN}(G,D) = \mathbb{E}_{x,y}[\log D(x,y)] \\ +\mathbb{E}_{x,z}[\log(1-D(G(x,z)))]\end{aligned} \tag{1}$$

The generator tries to minimize this function, while the discriminator tries to maximize it. y is the target, and $G(x,z)$ is the generated output image formed from an input image x and a random noise vector z where $G:\ \{x,z\} \rightarrow y$.

As shown in Fig. **??**, the generator **G** is trained to produce a visually enhanced version of the input image that represents the degradation, while the discriminator **D** is trained to distinguish the fake image produced by **G** from the real images. The **D** model is a pixel CNN-based classifier network that forces the **G** network to generate cleaner images by using the adversarial loss. In this way, the generator's loss is determined based on the discriminator's classification; if it successfully tricks the discriminator, it is rewarded, otherwise it is penalized. Note that for each iteration of GAN training, the discriminator is optimized first, followed by the generator through the backpropagation process and the objective loss function.

The $\mathcal{L}_1$ norm module derives the pixel-wise distance similarity between the fake image and the target image to allow the generator to produce less blurred images. In doing so, the $\mathcal{L}_1$ loss updates the generator's parameters using the assumed optimizer to improve its performance by achieving consistent gradients in the first iterations. The final GAN objective function can be defined as follows:

$$\acute{G} = \min_G \max_D \mathcal{L}_{cGAN}(G,D) + \lambda\mathcal{L}_1 \tag{2}$$

Where λ is a regularization parameter assigned to the $\mathcal{L}_1$ loss to improve the image generation quality. The GAN testing process allows only the trained generator to be used for inference, requiring low computational cost. In the following subsections, the **G** and **D** networks are introduced and their architecture and learning are discussed.

3.2 Pixel-wise discriminator

As mentioned earlier, the discriminator is a binary classifier that monitors the outputs of the generator and learns to distinguish these outputs from the targets. According to this strategy, classification can be performed

considering the whole image and assigning the corresponding label, or using non-overlapping regions with their class labels, or using pixel-level classification by analyzing each pixel individually instead of working at a global level. The pixel-based discriminator proved to be the most consistent method, providing the generator with a high gradient and significant adversarial loss for improved image generation and avoiding averaging phenomena [3]. The discriminator network **D** used in this work is based on a pixel classification architecture. As shown in Fig. **??**, it consists of a network of convolutional layers and activation layers that assign a binary label (fake/real) to each pixel. The **D** network has a rectilinear structure, since the first layers estimate the features, while the last layers create the classification map with two labels (0 = fake, 1 = real). It classifies both the real and fake data from the generator **G**. By maximizing the loss function in Equation 2, it penalizes itself if it classifies a real instance as fake or a fake instance (produced by the generator) as real. The optimization rule for training the discriminator **D** is given as follows:

$$\nabla_{\mathbf{D}} \frac{1}{m} \sum_{k=1}^{m} [log(\mathbf{D}(\tilde{I}_k)) + log(1 - \mathbf{D}(\mathbf{G}(I_k))) \tag{3}$$

Where $log(\mathbf{D}(\tilde{I}_k)$ indicates the loss in classifying the ground truth image $\tilde{I}$ as a real image, and $log(1 - \mathbf{D}(\mathbf{G}(I_k)$ measures the errors made by the discriminator in handling the fake image I and in being outwitted by the generator network.

3.3 The proposed MFNet generator

The generator part of a GAN learns to generate plausible data that has the same data distribution as the target data. This section introduces a new and effective deep CNN generator network called MFNet to improve the performance of GAN while motivating the discriminator to perform at its best to achieve better regularization and convergence learning speed. As shown in Fig. **??**, our MFNet block consists of two CNN modules, the feature extraction module $\mathbf{M_F}$ and the feature compression module $\mathbf{M_C}$, which are chained to connect both the downsampling encoder **E** and the upsampling decoder **D**. Autoencoders **E** and **D** are a special type of feedforward neural networks where the input is the same as the output. They compress the input into a low-dimensional latency space representation using the bottleneck layer and then reconstruct the output from this representation. However, the ability of autoencoders to perform automatic feature extraction is limited. Alternatively, we consider implementing autoencoders as part of our framework with the proposed MFNet block to achieve better feature extraction and compression. The encoder **E** generates low-resolution representations of the input image and learns to downsample it to a set of feature maps. In our framework (cf. Fig. **??**), the encoder receives the input images with size $l \times w \times c$ and downscales them to a feature representation (code) of size $L \times W \times C$ where L and W are significantly smaller than l and w and C is significantly greater than c. The encoder **E** consists of 4 convolutional blocks defined by convolution, batch normalization, and ReLU activation layer $[Conv, BN, ReLU]$. The convolution of the first three blocks has a kernel size of 3×3 (with stride =1) and 1×1 for the last block (with stride = 2). The encoded feature map is processed by the MFNet block for feature extraction and compression at various scales. Then, the MFNet output feature map is fed to the decoder **D** to project the feature-related information learned from the encoder to the higher pixel space resolution. By this, the decoder **D** upscales the output map of MFNet block through two transposed-based convolutional blocks with a transposed convolution with 3×3 kernel size, batch normalization, and ReLU $[TransConv, BN, ReLU]$. The last layer of the decoder is a transposed convolution with 3×3 matching the same kernel size of the first encoding block. This decoding layer does not include activation and normalization operations since it is the last one and our output is an image. Unlike the U-net generator, one of the main advantages of our MFNet block is that it can be used in plug-and-play mode. It is a generic block that supports any autoencoder architecture thanks to its adaptive layers and dynamic kernel filter mapping. The feature extraction module $\mathbf{M_F}$ learns to capture necessary and discriminative information from the output map of the encoder **E**, reflecting and characterizing structural and noise details related to the evaluated task. The overall performance of our GAN depends on the capabilities of the $\mathbf{M_F}$ module, as the extraction of meaningful features significantly improves the GAN performance.

As shown in Fig. **??**, the feature extraction module $\mathbf{M_F}$ branches into four CNN streams. The first CNN stream consists of four 2-D convolutional layers with filters of size 3×3 and 64 filters. It applies sliding convolution filters to the input map to compute the dot product of the weights and the input. Using convolutions with a filter size of 3×3 within this CNN stream helps the MFNet block to capture high-frequency details (e.g., image noise) and separate them from structural information in the image, as is done in image denoising. This avoids losing some high-frequency details while preserving the structural information of the im-

age, and by using multiple layers, the MFNet block becomes deeper to capture more complex features. The second CNN stream has two 2-D convolutional layers and a kernel size of 5×5. These convolutional filters take into account the close spatial relationship between adjacent pixels.

This could be particularly useful in many computer vision tasks, such as image demoireing, which often requires detection of moiré patterns where certain components have a specific spatial relationship to other components. The third CNN branch is a 2-D convolutional layer with a filter size of 7×7 to process low-frequency information of the image, as in the case of infrared image translation. In this task, the GAN network converts visible images into this kind of low brightness images, changing the structural information of the input image. Therefore, larger receptive fields of 7×7 are required to reconstruct the structural integrity of the image and process the low-frequency features that may look like part of the background. The fourth stream serves as an auxiliary branch that contains a single 2D convolutional layer with a filter size of 1×1 as a residual connection to reduce dimension in depth while retaining salient features for efficient low-dimensional embedding and feature pooling. The residual connection helps solve the problem of shattered gradients by providing spatial structure to the gradients. The combination of different filter sizes provides a spatially balanced feature distribution, emphasizing different aspects of the input map and more nonlinearity between CNN branches to deeply extract high and medium feature frequencies. This helps the MFNet block to effectively capture shape-variable and complex patterns (e.g., artifact and blur patterns) while preserving the high-frequency details.

The outputs of the four CNN streams are merged using a concatenation layer to form a holistic feature map with size $L \times W \times (4 \times 64)$ while maintaining all the discriminative details. The motivation to use multi-CNN branches as feature extraction is related to their experimental capabilities to capture discriminant and perceptual features at different scales thanks to the different kernel sizes. In the overall feature extraction module $\mathbf{M_F}$, the decrease in the number of 2-D convolutional layers is accompanied by an increase in the filter size. This is related to the assumption that a larger filter size captures more details than a smaller one. Therefore, we used fewer CNN layers when filter size is important to preserve time complexity per layer and reduce sensitivity to network initialization.

The feature compression module $\mathbf{M_C}$ (see Fig. **??**) compresses the obtained concatenated feature map and attempts to restore the same dimensional feature representation as the original output of the encoder. The module $\mathbf{M_C}$ contains the same CNN branches with the same number of layers per branch. The only difference between $\mathbf{M_F}$ and $\mathbf{M_C}$ is the number of filters, which is equal to the number of filters $\mathbf{(C)}$ of the encoder divided by the number of CNN branches $\mathbf{(C/4)}$. This compresses the corresponding features and spatially aligns the autoencoder with MFNet block. This CNN layout strategy makes the MFNet block a generic model that can work with all autoencoders.

As previously stated, the optimization of the MFNet generator is based on two losses. The first loss is the L1-norm, which measures how close the generated image is to the target image. The real/fake loss calculated by the discriminator $\mathbf{D}$, as expressed in Equation 3, is the second loss for optimizing the generator $\mathbf{G}$. This loss indicates how much $\mathbf{G}$ has tricked $\mathbf{D}$, and shows that $\mathbf{G}$ is trying to provide images that are as close as possible to the target images. If the generated image contains more pixels with real distribution than with fake distribution, the real/fake loss is reduced. $\mathbf{G}$ is induced to compute a more realistic image than in the previous iteration because $\mathbf{D}$ is optimized to detect the fake image. The optimization rule for training the generator $\mathbf{G}$ is given as follows:

$$\nabla_{\mathbf{G}} \frac{1}{m} \sum_{k=1}^{m} [log(\mathbf{D}(\mathbf{G}(I_k))) + \lambda |\tilde{I}_k - \mathbf{G}(I_k)|_1] \tag{4}$$

Where $log(\mathbf{D}(\mathbf{G}(I_k)))$ is the loss of classifying the pixels of the generated image I as real ones, and $\lambda|\tilde{I}_k - \mathbf{G}(I_k)|_1$ is the L1-loss with a regularization parameter λ.

4 Experiments

To validate and demonstrate the effectiveness of the proposed framework, we evaluate it on five challenging image translation and enhancement applications. In this section, we describe the tested benchmarks with their evaluation setup, present a comprehensive experiment to demonstrate the accuracy and efficiency of the proposed MFNet generator compared to current literature generators, and report the quantitative image generation performance compared to SOTA approaches.

4.1 Performance comparison with the baseline generators

We have conducted extensive experiments on the following five challenging and widely studied image translation and enhancement tasks:

- **Image Demoireing**. The aim of this application is to recover clean versions of images damaged by a variety of moiré degradations, usually caused by interference between the camera's pixel grid and the screen of an electrical device. TIP2018 [28] is the selected benchmark, which includes 135,000 image pairs, each consisting of an image damaged by moiré degradation and an uncorrupted reference image. In the experiments, we setup the standard protocol where 90% of the data is used for training and 10% for testing.
- **Visible-to-Infrared**. This image translation task consists of converting RGB images into synthetically generated infrared images. GAN -based approaches are a practical solution for such types of low-light image translation. In experiments, we have tested the proposed method using the challenging LLVIP benchmark [26], which contains 15488 image pairs (visible and infrared) with different scenes acquired under different low-light conditions. In our evaluations, the standard protocol is applied, using 3463 image pairs for testing and the remaining data for training.
- **Underwater Image Enhancement**. Images captured underwater are often distorted by absorption and scattering effects caused by the physical characteristics of the underwater environment, including lighting conditions, water type, and depth. This image degradation results in low contrast, color shifts, and like- haze patterns. Several approaches to enhance underwater images have been presented in the literature, most of which are based on different variants of GAN. In our experiments, we selected the image enhancement benchmark (UIEBD) [36] for testing, which consists of 890 images affected by underwater degradation. Note that the current SOTA trained their CNN-based models on a different dataset of 6128 image pairs (clean and with water haze) generated using CycleGAN [37]. Because only synthetic reference images are available in the UIEBD dataset, two non-reference assessment metrics were used in the SOTA approaches: the underwater color image quality evaluation (UCIQE) [52] and the underwater image quality measure (UIQM) [53]. For both metrics, the higher the value, the better the image enhancement quality.
- **Image Denoising**. It deals with the high frequency details in the image consisting of Gaussian noise patterns. The task is to clean up the noisy image by separating these unwanted patterns from the structural information of the image. The proposed MFGAN approach is tested on the color CBSD68 dataset [41]. For training our model, we used the training set of the single image super resolution dataset DIV2K [54] and the BSDS500 [55] dataset as arranged in [56].
- **JPEG Artifact Removal**. JPEG artifact removal is a challenging computer vision task focused on restoring and enhancing images that have been compressed using the Joint Photographic Experts Group (JPEG) compression algorithm. This image compression can introduce unwanted artifacts, which significantly impact the visual quality of the image. These artifacts manifest in the form of blocky patterns, blurring, or curling at the edges. The degree of degradation of JPEG images is determined by the quality factor (QF), which is a compression factor between 0 and 100. Higher values of QF correspond to less loss of image detail and better overall quality. In our experiments, we focused on three quality factors (QFs): 10, 20, and 30, which are commonly used in the current state of the art. We applied the OpenCV compression function imencode() to the Div2K training dataset [54] to create the JPEG training data. Our MFGAN is evaluated on two widely used benchmark datasets, namely the Live1 and Classic5 test sets.

The proposed MFNet block is a generic feature extractor that supports all types of autoencoders thanks to its adaptive CNN convolution and dynamic feature assignment. To prove the reliability and superiority of the MFNet generator, a comprehensive evaluation is presented in this section to compare its performance with the standard ResNet and U-Net generators. Note that we evaluate MFNet generator using the same GAN architecture presented in Section 3, which includes the pixel-wise discriminator and the autoencoder used. Except for the dewatering application, which uses non-referential metrics for evaluations, we validated the performance of the baseline generators along with our proposed generator using two widely used image quality metrics: Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM).

Image quality is evaluated through visual analysis (see Figure **??**) of the generated image, ground truth (GT), and original input. The PSNR, SSIM, UCIQE, and UQIM values recorded by the proposed MFNet generator, as well as the ResNet6, ResNet9, and U-Net generators on the five tested applications, are listed in Tables **??** and **??**. As can be seen, the MFNet generator outperforms all the base generators for every application tested. In the TIP2018 dataset, MFNet achieves a high PSNR/SSIM of 31.64/0.961, followed by U-Net, ResNet6, and ResNet9 with 29.85/0.9428,

28.74/0.9417, and 28.56/0.9343, respectively. MFNet's high quantitative performance in the above tasks proves its effectiveness in recovering clean images thanks to its multi-kernel CNN filtering to capture different degradation patterns. Such patterns are filtered from structural information in the image, and by using multiple CNN layers, the MFNet generator becomes deeper to handle more complex features. This high quantitative performance of MFNet in image demoireing is confirmed by the visual results given in the 2nd row of Figure **??**. Our MFNet successfully removes moiré patterns and corrects contrast and color shift, while preserving the structural details of the image and ensuring good quality image generation that is similar to the ground truth image (GT). The U-Net generator also succeeds in generating a good fake image, but fails to completely remove the yellow color distortion from the input. The image generated by ResNet9 still suffers from moiré and color distortion patterns.

In the LLVIP dataset, MFNet also ranks first with {PSNR:21.54, SSIM:0.6995}, outperforming U-Net, ResNet6, and Res-Net9 by {PSNR: +1.46 dB, SSIM:+5.26%}, {PSNR: +3.09 dB, SSIM: +5.78%}, and {PSNR: +5.06 dB, SSIM: +24.55%}, respectively. Compared to the U-Net and ResNet generators, the high performance of MFNet is achieved thanks to the feature extraction and compression modules, which provide discriminant feature transformation from visible to infrared space. The fact that the U-Net architecture does not have a feature extraction block limits this transformation to perform better image translation. On the other hand, the ResNet variants are not deep enough with respect to the CNN layers to ensure good feature extraction. The promising quantitative results of our proposal can be verified by the qualitative evaluation in Figure **??**, 1st row, where MFNet achieves better visual improvements that closely match the ground truth (GT), in comparison to U-Net. Indeed, pedestrians and cars are barely visible in low-light conditions, but they are easy to detect and outline in the infrared images generated by our MFNet. The quality of the infrared images generated by the U-Net generator is reasonably good, but the outlines of cars, objects, and pedestrian areas are blurred and not clear enough, and there is less internal texture information in the image. The ResNet9 produces poor image quality because the background and car regions are incorrectly translated to infrared, but it manages to distinguish people better than the U-Net generator. In the TIP2018 and LLVIP datasets, U-Net was able to achieve better PSNR and SSIM scores compared to the ResNet variants. In the image demoiering task, the U-Net generator manages in ensuring feature reusability between the degraded image and the generated image by discarding only the moiré patterns while preserving the structural details. This is thanks to its skip connection between upconvolution and downconvolution, which reuses some structural features of the original image. In the LLVIP dataset, the good performance of U-Net compared to ResNet variants can be justified by the fact that converting visible images to infrared images is more challenging than image enhancement, which requires more learnable parameters and greater depth to fully convert RGB data to infrared data.

In the CBSD68, Live1, and Classic5 datasets, the MFNet generator demonstrates superior performance compared to the base generators, offering a reliable solution for effectively removing Gaussian noise (CBSD68) and JPEG degradation (Live1 & Classic5) from images with high accuracy in terms of PSNR and SSIM. It can also be seen that as the degradation level in the images gradually increases, the performance of the tested generators decreases. This observation is reasonable since the task of image denoising becomes progressively more challenging as the model attempts to separate noise patterns entirely from the image. Consequently, the generated image may lose some structural details as the noise level increases. Likewise, in image deblocking, as the level of JPEG degradation increases, the compressed images contain more blocky artifacts and less high frequency information. This complicates the learning process for the generators and hinders the generation of realistic images, as they must compensate for the loss of fine details and handle the blocky artifacts. In both image denoising and deblocking, the ResNet variants quantitatively perfrom better than U-Net generator. This can be attributed to the ResNet variants' use of convolutions with a filter size of 3×3, which effectively capture the noise and JPEG degradation information as high frequency details. In contrast, the U-Net generator lacks this capability. These statements can be visually verified in Figure **??**, rows 4th, 5th, and 6th. For image denoising (cf. 4th row), it can be clearly seen that MFNet highly recovers both the structural information (e.g., the shape of the star pattern in the image) and the texture details, closely matching the ground truth (GT) image. The quality of image generation by ResNet9 is better compared to U-Net, as it generates a clean image but does not correctly reconstruct some details such as the star shape. Image generation with U-Net still results in noisy degradation with blurring effects around the star pattern. In image deblocking, it can be seen that MFNet generator perfectly restores color tone and other visual features such as lighting and reflection with respect to the ground truth (GT) of the Classic5 dataset

(see 5th row). Moreover, the output of MFNet exhibits smoother patterns and finer transitions and edges compared to those generated by ResNet9 and U-Net. Moving on to the Live1 dataset (see 6th row), the U-Net output produces blurred and distorted patterns around and within the cropped text box, which compromises its overall quality. On the other hand, the ResNet output manages to enhance the image visually when compared to U-Net. However, it introduces some false and blurred texture details in some regions and slightly generates color shifts in the cropped text box. In contrast, the MFNet-generated image demonstrates visual improvements in various aspects. It presents clearer contours of the text box, effectively removes blocky and blurring patterns, and successfully recovers intricate structural details when compared to the baseline generators. It is worth noting that using a JPEG quality factor of 10, which is the most severe and challenging degradation level, still results in some artifacts that are visible in the background of the generated images in the Live1 dataset.

For image dewatering, MFNet achieves the highest performance of {UCIQE: 0.6992, UQIM:6.4291}, outperforming ResNet6, ResNet9, and U-Net generators by {UCIQE: +0.16, UQIM: +1.3678}, {UCIQE:+0.1856, UQIM:+2.0545}, and {UCIQE:+0.2005, UQIM:+2.5275}, respectively. This quantitative performance can be verified using the subjective evaluation in Figure **??**, 3rd row. Compared to the U-Net and ResNet9 methods, the proposed MFNet generator corrects the color distortion and brightness of the input and tries to reach the ground truth (GT) image with less blue tones while preserving the image details. On the other hand, the image generated by U-Net reduces like-haze patterns but affects the blue-green effects and leads to deterioration of the color shift in the bluish water region. The ResNet9 generator retains the blue-green effects but fails to correct the image illumination. The promising result obtained by our proposal in this dataset can be attributed to the CNN capability of MFNet, as it provides a combination of different filter sizes that help the overall GAN framework to spatially balance the feature distribution and emphasize high and medium feature frequencies such as color casts, low contrasts, and haze-like underwater degradation effects.

4.2 Performance comparison with the literature

This section presents a comprehensive performance evaluation conducted to compare our proposed MFNet-based GAN with current state-of-the-art approaches. Tables **??**, **??**, **??**, and **??** illustrate the quantitative results obtained on the five targeted image translation and enhancement tasks, along with the results of the well-known state-of-the-art approaches. In this section, we refer to our MFNet-based GAN as MFGAN for simplicity. From the three tables, the proposed MFGAN method has achieved the best performance and guarantees higher evaluation metrics compared to SOTA approaches.

In the image demoireing task (TIP2018 dataset), the MFGAN method outperforms the state-of-the-art with PSNR/SSIM values of 31.64/0.961, as shown in Table **??**. The ESDNet approach proposed in [57] is the second best system with a PSNR /SSIM value of 30.11/0.920, which is outperformed by our MFGAN by +1.53 dB/ 4.1%. The technique in [57] consists of a semantic aligned scale-aware module specifically introduced to address the scale variation of moiré patterns. The third best SOTA system is the MBCNN method [35], which consists of a special loss, the Advanced Sobel Loss (ASL), and a multiscale bandpass convolutional network proposed specifically for image demoireing. In terms of network parameters, the MBCNN method is very time consuming and requires a significant number of learnable network parameters, as reported in their work in [35]. They introduced a light version of MBCNN with fewer parameters, but achieved less performance in removing moiré patterns from images. In [34], all the core modules of the moiré pattern removal neural network, known as MopNet, were specifically designed for the unique properties of moiré patterns, including a complex combination of loss functions consisting of the weighted pixel-wise Euclidean distance and the feature-based metric for training the MopNet network. However, it showed limitations in some scenarios where some output images contain irregular textures such as gravel or pavement in the background, as shown by the qualitative performance in [34]. This results in low quantitative performance with a PSNR of 27.75 and an SSIM of 0.895, surpassed by our MFGAN with a gain of +2.81 dB/5.8%. The work of [58] proposed a ushaped transformer and improved its objective function by using a complex loss metric known as Charbonnier, but its performance is still competitive compared to our proposal with a PSNR/SSIM gain of +1.28 dB/3.6%. In terms of network complexity, our MFGAN model has 44 million parameters, which is less complex compared to Uformer-B model with 50.88 million parameters. FHDeNet [59], short for Full High Definition Demoireing Network, is a method developed for effectively removing moiré patterns from images. It utilizes a strategy of global to local pattern removal and employs a frequency-based high-resolution content separation mechanism to improve the image demoireing

task. The EDSR [60] is a ResNet -based approach originally introduced for single-image super-resolution task. They modified ResNet model by removing unnecessary modules in conventional residual layers to construct the EDSR. The same technique was re-implemented by [35] to extend its evaluation for image demoireing.

For the LLVIP dataset (see Table **??**), MFGAN provides a reliable solution for synthesizing visible images into infrared images with a PSNR/SSIM accuracy of 21.54/0.70, which is a significant performance improvement of +9.35 dB/43% and +10.78 dB/52% over the Pyramid-Pix2Pix and Pix2Pix GAN models, respectively. The latter achieved quantitatively low performance and qualitatively fails to produce correct infrared images. As reported in [26], the background in the generated image is blurred and the contours of the pedestrian and vehicle shapes are cluttered, filled with false details, and interspersed with artifacts. This is due to the use of the U-Net baseline generator model, which has its limitations in many image enhancement tasks and suffers from various convergence problems due to the lack of feature extraction blocks. The Pyramid-Pix2pix [27] is an extension of pix2pix architecture by incorporating a pyramid structure. This means that instead of operating solely at the original image resolution, Pyramid-Pix2pix works with multiple scales or resolutions of the input image to capture information at different levels of detail. LLVIP dataset poses a major challenge because infrared images are acquired through the temperature field of the object surface, which makes it difficult to fully discriminate objects in low-light conditions. Therefore, there is still room for improvement to accurately translate visible images into infrared images for low-light vision applications.

In the dewatering experiment (see Table **??**), the recorded UCIQE and UQIM results demonstrate the superiority of the proposed MFGAN in handling low contrast, color shift and haze-like underwater degradation, as it achieves the highest SOTA values of {UCIQE: 0.6992, UQIM: 6.4291}. The task-specific ACCE-D approach [61] is the second best system in this dataset with a UCIQE score of 0.6543, outperformed by our MFGAN by 4.49%. The ACCE-D framework employs the Difference of Gaussian (DoG) filter and bilateral filter to decompose the high-frequency and low-frequency details. It aims to suppress underwater noise specifically in the high-frequency component. The authors reported many problems related to the algorithm's complexity, including a large number of iterations and parameters that needed optimization. The DewaterNet [40] and TOPAL [62] models use a multi-term adversarial objective norm combining different dewatering-related loss functions, while in our approach we only use the simple L1 norm to optimize the generator parameters. The WSCT [39], FGAN [38], UWCNN [63], and UGAN [37] approaches also use multi-term objective functions to correct underwater degradation, but these methods introduce color shifts in some challenging scenes as indicated in their work. The BVS [64] is built upon a novel dehazing mechanism that incorporates two pathways: the structure pathway and the texture pathway. This approach aims to address the contrast and brightness attenuation present in underwater images.

In the CBSD68 color dataset, MFGAN's performance consistently outperforms all state of the art denoisers across all noise levels. This promising result can be attributed to MFGAN's powerful and effective multi-kernel prior modeling capability. Our solution perfectly recovers clean versions of images affected by noise degradation with high PSNR values of 35.21, 35.15, and 34.84, outperforming the second ranked SOTA approach ADL [65] by +0.6 dB, +2.97 dB, and +5.82 dB for $\sigma = 15$, $\sigma = 25$, and $\sigma = 50$ respectively. The ADL model, proposed in [65], is a denoising GAN that combines the design architecture of Efficient and Unet networks to effectively denoise images. For the noise level of $\sigma = 75$, MFGAN achieves a PSNR value of 34.25, surpassing the second-ranked approach, AdaFM-Net [66], by +7.9 dB. AdaFM-Net is constructed with a novel Adaptive Feature Modification (AdaFM) layer, which modifies middle-layer features using depthwise convolution filters. This layer enables continuous modulation of restoration levels within a single CNN for image restoration. Given the competitive performance of the ADL method over $\sigma = 15$, $\sigma = 25$, and $\sigma = 50$, it should be mentioned that it relies on a complex combination of specific noise loss functions, whereas we have used only the simple L1 loss to measure how far the model generation deviates from the ground truth. In terms of network parameters, the ADL denoiser requires 4.75 million parameters, which is much less complex than the MFGAN and IPT [67] methods. However, it is known that network parameters is not always considered as an important performance indicator, and a few SOTA approaches have reported the network complexity of their models.

The Image Processing Transformer (IPT), introduced in [67], used multi-head and multi-tail modules, along with a contrastive learning strategy, to address various low-level computer vision tasks such as denoising, super-resolution, and de-raining. The PaCNet [68] approach consists of augmenting video sequences with patch-craft frames and feeding them to a spatial CNN network to enhance the denoising task. In the LIDIA [69] approach, the authors employed a combination of

supervised and unsupervised techniques. The supervised technique aims for a universal denoiser, while the unsupervised technique focuses on instance adaptation. The overall framework leverages patch processing, nonlocal self-similarity, representation sparsity, and a multiscale treatment to improve the image denoising task. El Helou and Süsstrunk proposed the CBUIFD75 approach in [70], which consists of a theoretically-grounded blind and universal deep learning image denoiser for additive Gaussian noise removal. It uses a Bayesian technique to effectively estimate and remove noise from the image. Dror and Elad [71] presented novel insights into the convolutional sparse coding model and its ability to effectively represent natural images. To improve image denoising, they introduced a feed-forward CNN architecture in which each layer corresponds to a specific step in the sparse-coding processing flow. The RIDNet [72], short for Real Image Denoising Network [72], is a CNN-based denoising model that integrates a feature attention mechanism. This mechanism enables the rescaling of channel-wise features while considering the inter-channel dependencies present in the image. The Spatial-CNN presented in [73] uses spatial and temporal information in a feed-forward process to effectively denoise the images.

In the deblocking experiment, we conducted a comparative performance evaluation using three commonly used JPEG QF (Quality Factor) values, as shown in Table **??**. These values were selected based on their prevalence in the literature. We employed QF = 30 to represent low degradation, QF = 20 to represent medium degradation, and QF = 10 to represent high degradation. Although some studies include QF = 40, we chose not to include it in our evaluation because the degradation at this level is minimal and the quality of the input image can be easily restored. In accordance with the state-of-the-art assessment protocol, we reported the SSIM metric as it better reflects reconstruction quality compared to the PSNR metric. This choice is justified by the fact that the majority of state-of-the-art methods provided only the SSIM scores, making it a more suitable and commonly used metric for evaluation. In the following, we provide a brief overview of state-of-the-art (SOTA) methods for image deblocking, which are referenced in Table **??**. The D2SD method, introduced by Liu et al. [74], employs a data-driven approach to soft decoding in both the DCT and pixel domains. By leveraging residual redundancies in JPEG code streams and exploiting sparsity properties of latent images, it effectively restores JPEG-compressed images. Zhao et al. [75] developed the SSRQC algorithm, which combines structural sparse representation (SSR) prior with quantization constraint (QC) prior to reduce JPEG compression artifacts. The SSR prior effectively enforces local sparsity and non-local self-similarity of images, while the inclusion of QC ensures a more reliable and robust estimation process. Chen et al. [76] proposed the DPW-SDNet, a deep CNN-based soft decoding network operating in the dual pixel-wavelet domains to restore the compressed images. Suna et al. [77] introduced CARD, a network designed specifically for reducing JPEG compression artifacts. It captures the nonlinear mapping relationship between the DCT coefficients of the compressed image and those of the original image, using the information in the DCT domain for effective image restoration. The comprehensive overview of the ARCNN [48], TNRD [49], DnCNN-3 [30], and ARN [50] methods can be found in the related work section. From the listed results in Table **??**, it can be observed that MFGAN consistently outperforms all state-of-the-art approaches in the Classic5 and Live1 datasets across all JPEG degradation levels. This performance can be primarily attributed to the convolutional feature extraction capability of the MFNet generator. By integrating different filter sizes, MFNet effectively facilitates the spatial balance of feature distribution within the overall GAN. Moreover, MFNet's CNN architecture emphasizes high and medium feature frequencies, enabling the successful mitigation of JPEG degradation artifacts while preserving structural details in the image. This combination of spatial compensation and frequency emphasis is a key factor in the performance demonstrated by our approach.

The above extensive experiments demonstrate the reliability and effectiveness of the MFGAN model in improving the quality of image generation for various applications. From the evaluation of the SOTA performance comparison, it can be concluded that relying only on the design of task-specific loss functions and paying less attention to the architectural generator modules is not always sufficient to achieve better performance. This justifies our motivation to design a new and conceptually simple generator model that extracts more discriminant and conditional features to improve image restoration tasks. Unlike most SOTA approaches that use complex and task-specific combinations of different loss functions, our solution uses only the simple L1 norm to update the model weights. Another advantage of the proposed MFGAN is the multi-kernel filtering of CNN streams to capture high-to-low feature frequencies reflecting various shape-variable and complex image degradations while preserving the structural image details. This modeling strategy makes the overall framework a generic solution for any type of image generation and enhancement application.

4.3 Ablation study

In this section, we present an ablation study in which a series of experiments were performed on the benchmarks tested to investigate the effectiveness and performance stability of our various system components. Table **??** summarizes the results of the ablation study for the following system configurations. First, as shown in Table **??**, we evaluated the GAN architecture, denoted $[(Encoder + (SS-N-F \times F) + Decoder) + Dis]$, which consists of the encoder, the $MFNet$ block $(SS-N-F \times F)$ containing a single CNN stream of N convolutional layers with filters of size $F \times F$, followed by the decoder and the GAN discriminator (Dis). Following this architectural configuration, we tested $MFNet$ settings $(SS-4-3\times3)$, $(SS-2-5\times5)$, and $(SS-1-7\times7)$. Note that each of these individual CNN streams, as shown in Fig. **??**, is used in both the feature extraction module $\mathbf{M_F}$ and the feature compression module $\mathbf{M_C}$. These two modules are chained together to connect the downsampling encoder and the upsampling decoder. In the second configuration, we evaluated the MFGAN architecture without the CNN auxiliary branch, which contains a single 2D convolutional layer with a filter size of 1×1. The third configuration consists of the full MFGAN architecture as shown in Fig. **??**.

Table **??** shows that the full MFGAN architecture $(Cfg3)$ outperforms all other GAN architectures in all benchmarks tested. It achieves the highest performance with PSNR/SSIM: 31.64/0.961 (TIP2018), PSNR/SSIM: 21.54/0.70 (LLVIP), SSIM:0.8389 (Live1 in QF10), SSIM:0.9267 (Live1 in QF30), PSNR:35.21 (CBSD-68 in $\sigma = 15$), and PSNR:34.25 (CBSD68 in $\sigma = 75$). In the UIEBD dataset, $MFGAN - Cfg3$ also ranks first with the highest score of {UCIQE: 0.6992, UQIM: 6.4291}. In the CBSD68 and Live1 datasets, it can be seen that the performance results are better at noise level 15 (CBSD68) and JPEG QF30 (Live1) compared to noise level 75 and JPEG QF10. This outcome is expected since the models aim to eliminate the severe noise (level 75) and high blocky JPEG artifacts (QF10) that could potentially impact the structural details of the generated image. As shown in Table **??**, adding the auxiliary branch on $MFGAN - Cfg2$ to construct the full $MFGAN - Cfg3$ (cf. Fig. **??**) improves the overall performance in all tested datasets. The auxiliary branch consists of a CNN residual connection with a filter size of 1×1 to give spatial structure to the gradients while maintaining salient features for efficient low-dimensional embedding and feature pooling. This helps the overall model avoid the problem of shattered gradients and improves its performance in image generation. The $Cfg1-2$ model achieves the third-best performance in the CBSD68, Live1, and UIEBD datasets and the fourth-best performance in the TIP2018 and LLVIP datasets. The same performance behavior is observed for the $Cfg1 - 3$ model. This demonstrates the instability of performance when using single CNN streams to recover clean versions of degraded images. In contrast, our full MFGAN configuration experimentally demonstrates its effectiveness in image restoration thanks to its multi-kernel CNN streams for filtering different degradation patterns covering high to low feature frequencies. This combination of different CNN streams ensures a spatially balanced feature distribution that characterizes shape-variant and complex image degradations while preserving structural image details.

5 Conclusion

This paper introduced a generic GAN called Multi-Kernel Filter-based Conditional Generative Adversarial Network (MFGAN) to address various challenges in image restoration and translation tasks. This is achieved by proposing a novel generator architecture to learn more discriminative and perceptual features that reflect image degradation patterns and structural image details. The network structure of our generator consists of two CNN modules, feature extraction, and feature compression, each branching into four CNN streams with multiple layers at different scales to deeply capture shape-variant and complex patterns while preserving the low-frequency details that characterize the structural integrity of the image. This multi-kernel filtering strategy makes our FMGAN a generic solution for any image generation and enhancement application. Moreover, the proposed generator is a flexible network since it can be spatially aligned with any type of autoencoder architecture thanks to the adaptive CNN convolution and dynamic feature assignment. A comprehensive evaluation was conducted to compare our MFNet generator with baseline generators and demonstrate the reliability and superiority of our proposal in generating visually appealing results with high quantitative performance. Compared to SOTA approaches, the proposed MFGAN method has achieved the best performance and guarantees higher evaluation metrics for all tested image restoration and translation tasks, including image demoireing, visible-to-infrared image translation, underwater image enhancement, JPEG artifact removal, and image denoising. We also note that unlike most SOTA approaches that use multiple task-specific loss functions, the network weights of our MFGAN were updated using only the simple L1 norm.

In future work, it will be valuable to extend the evaluation of MFGAN to other challenging image generation tasks, such as image desnowing, dehazing, and deraining. The proposed MFGAN can also be used as a data augmentation technique to simulate environmental conditions where real-world data acquisition may be complex, e.g., nighttime imagery, or in rain, snow, fog, etc. In this way, sufficient synthetic data will be provided to train environmental perception models for autonomous vehicles or video surveillance. Finally, instead of the traditional GAN loss functions, we plan to define a generic adaptive loss function that will be trained, updated, and optimized to match the component features for each image generation task tested.

Data Availability Statements

Data sharing will be made available on reasonable request.

References

1. Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Proceedings of the International Conference on Neural Information Processing Systems (NIPS 2014)*, page 2672–2680, 2014.
2. Yoav Shoham. Game theory pragmatics: A challenge for AI. In *Proceedings of the Twenty-Third AAAI Conference on Artificial Intelligence, AAAI 2008, Chicago, Illinois, USA, July 13-17, 2008*, pages 1606–1608. AAAI Press, 2008.
3. Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A Efros. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1125–1134, 2017.
4. Chaoyue Wang, Chang Xu, Chaohui Wang, and Dacheng Tao. Perceptual adversarial networks for image-to-image transformation. *IEEE Transactions on Image Processing*, 27(8):4066–4079, 2018.
5. Nilesh Pandey and Andreas Savakis. Poly-gan: Multi-conditioned gan for fashion synthesis. *Neurocomputing*, 414:356–364, 2020.
6. Jun-Yan Zhu, Philipp Krähenbühl, Eli Shechtman, and Alexei A Efros. Generative visual manipulation on the natural image manifold. In *European conference on computer vision*, pages 597–613. Springer, 2016.
7. Zhenliang He, Wangmeng Zuo, Meina Kan, Shiguang Shan, and Xilin Chen. Attgan: Facial attribute editing by only changing what you want. *IEEE transactions on image processing*, 28(11):5464–5478, 2019.
8. Christian Ledig, Lucas Theis, Ferenc Huszár, Jose Caballero, Andrew Cunningham, Alejandro Acosta, Andrew Aitken, Alykhan Tejani, Johannes Totz, Zehan Wang, et al. Photo-realistic single image super-resolution using a generative adversarial network. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4681–4690, 2017.
9. Carl Vondrick, Hamed Pirsiavash, and Antonio Torralba. Generating videos with scene dynamics. *Advances in neural information processing systems*, 29:613–621, 2016.
10. Mengyu Chu, You Xie, Jonas Mayer, Laura Leal-Taixé, and Nils Thuerey. Learning temporal coherence via self-supervision for gan-based video generation. *ACM Transactions on Graphics (TOG)*, 39(4):75–1, 2020.
11. Nannan Wang, Wenjin Zha, Jie Li, and Xinbo Gao. Back projection: An effective postprocessing method for gan-based face sketch synthesis. *Pattern Recognition Letters*, 107:59–65, 2018.
12. Xuanyi Dong, Yan Yan, Wanli Ouyang, and Yi Yang. Style aggregated network for facial landmark detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 379–388, 2018.
13. Grigory Antipov, Moez Baccouche, and Jean-Luc Dugelay. Face aging with conditional generative adversarial networks. In *2017 IEEE international conference on image processing (ICIP)*, pages 2089–2093. IEEE, 2017.
14. Karim Armanious, Chenming Jiang, Marc Fischer, Thomas Küstner, Tobias Hepp, Konstantin Nikolaou, Sergios Gatidis, and Bin Yang. Medgan: Medical image translation using gans. *Computerized Medical Imaging and Graphics*, 79:101684, 2020.
15. Hajar Emami, Majid Moradi Aliabadi, Ming Dong, and Ratna Babu Chinnam. Spa-gan: Spatial attention gan for image-to-image translation. *IEEE Transactions on Multimedia*, 23:391–401, 2020.
16. Zili Yi, Hao Zhang, Ping Tan, and Minglun Gong. Dualgan: Unsupervised dual learning for image-to-image translation. In *Proceedings of the IEEE international conference on computer vision*, pages 2849–2857, 2017.
17. Wallace Lira, Johannes Merz, Daniel Ritchie, Daniel Cohen-Or, and Hao Zhang. Ganhopper: Multi-hop gan for unsupervised image-to-image translation. In *European Conference on Computer Vision*, pages 363–379. Springer, 2020.
18. Xiaolong Wang and Abhinav Gupta. Generative image modeling using style and structure adversarial networks. In *European conference on computer vision*, pages 318–335. Springer, 2016.
19. Jun-Yan Zhu, Taesung Park, Phillip Isola, and Alexei A Efros. Unpaired image-to-image translation using cycle-consistent adversarial networks. In *Proceedings of the IEEE international conference on computer vision*, pages 2223–2232, 2017.
20. Phillip Isola, Jun-Yan Zhu, Tinghui Zhou, and Alexei A Efros. Image-to-image translation with conditional adversarial networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1125–1134, 2017.
21. Da-Wei Jaw, Shih-Chia Huang, and Sy-Yen Kuo. Desnowgan: An efficient single image snow removal framework using cross-resolution lateral connection and gans. *IEEE Transactions on Circuits and Systems for Video Technology*, page 1–1, 2020.
22. He Zhang, Vishwanath Sindagi, and Vishal M Patel. Image de-raining using a conditional generative adversarial network. *IEEE transactions on circuits and systems for video technology*, 30(11):3943–3956, 2019.
23. Huy Phan, Ian V McLoughlin, Lam Pham, Oliver Y Chén, Philipp Koch, Maarten De Vos, and Alfred Mertins. Improving gans for speech enhancement. *IEEE Signal Processing Letters*, 27:1700–1704, 2020.
24. Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In

Proceedings of the IEEE conference on computer vision and pattern recognition, pages 770–778, 2016.
25. Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*, pages 234–241. Springer, 2015.
26. Xinyu Jia, Chuang Zhu, Minzhen Li, Wenqi Tang, and Wenli Zhou. Llvip: A visible-infrared paired dataset for low-light vision. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 3496–3504, 2021.
27. Shengjie Liu, Chuang Zhu, Feng Xu, Xinyu Jia, Zhongyue Shi, and Mulan Jin. Bci: Breast cancer immunohistochemical image generation through pyramid pix2pix. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1815–1824, 2022.
28. Yujing Sun, Yizhou Yu, and Wenping Wang. Moiré photo restoration using multiresolution convolutional neural networks. *IEEE Transactions on Image Processing*, 27(8):4160–4172, 2018.
29. Jiwon Kim, Jung Kwon Lee, and Kyoung Mu Lee. Accurate image super-resolution using very deep convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 1646–1654, 2016.
30. Kai Zhang, Wangmeng Zuo, Yunjin Chen, Deyu Meng, and Lei Zhang. Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising. *IEEE transactions on image processing*, 26(7):3142–3155, 2017.
31. Li Xu, Qiong Yan, Yang Xia, and Jiaya Jia. Structure extraction from texture via relative total variation. *ACM transactions on graphics (TOG)*, 31(6):1–10, 2012.
32. Xi Cheng, Zhenyong Fu, and Jian Yang. Multi-scale dynamic feature encoding network for image demoiréing. In *2019 IEEE/CVF International Conference on Computer Vision Workshop (ICCVW)*, pages 3486–3493. IEEE, 2019.
33. Xun Huang and Serge Belongie. Arbitrary style transfer in real-time with adaptive instance normalization. In *Proceedings of the IEEE international conference on computer vision*, pages 1501–1510, 2017.
34. Bin He, Ce Wang, Boxin Shi, and Ling-Yu Duan. Mop moire patterns using mopnet. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2424–2432, 2019.
35. Bolun Zheng, Shanxin Yuan, Gregory Slabaugh, and Ales Leonardis. Image demoireing with learnable bandpass filters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 3636–3645, 2020.
36. Chongyi Li, Chunle Guo, Wenqi Ren, Runmin Cong, Junhui Hou, Sam Kwong, and Dacheng Tao. An underwater image enhancement benchmark dataset and beyond. *IEEE Transactions on Image Processing*, 29:4376–4389, 2019.
37. Cameron Fabbri, Md Jahidul Islam, and Junaed Sattar. Enhancing underwater imagery using generative adversarial networks. In *2018 IEEE International Conference on Robotics and Automation (ICRA)*, pages 7159–7165. IEEE, 2018.
38. Hanyu Li, Jingjing Li, and Wei Wang. A fusion adversarial underwater image enhancement network with a public test dataset. *arXiv preprint arXiv:1906.06819*, 2019.
39. Chongyi Li, Jichang Guo, and Chunle Guo. Emerging from water: Underwater image color correction based on weakly supervised color transfer. *IEEE Signal Processing Letters*, 25(3):323–327, 2018.
40. Hanyu Li and Peixian Zhuang. Dewaternet: A fusion adversarial real underwater image enhancement network. *Signal Processing: Image Communication*, 95:116248, 2021.
41. David Martin, Charless Fowlkes, Doron Tal, and Jitendra Malik. A database of human segmented natural images and its application to evaluating segmentation algorithms and measuring ecological statistics. In *Proceedings Eighth IEEE International Conference on Computer Vision. ICCV 2001*, volume 2, pages 416–423. IEEE, 2001.
42. Xiaojiao Mao, Chunhua Shen, and Yu-Bin Yang. Image restoration using very deep convolutional encoder-decoder networks with symmetric skip connections. *Advances in neural information processing systems*, 29:2802–2810, 2016.
43. Kai Zhang, Wangmeng Zuo, Shuhang Gu, and Lei Zhang. Learning deep cnn denoiser prior for image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 3929–3938, 2017.
44. Venkataraman Santhanam, Vlad I Morariu, and Larry S Davis. Generalized deep image to image regression. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pages 5609–5619, 2017.
45. Pengju Liu, Hongzhi Zhang, Kai Zhang, Liang Lin, and Wangmeng Zuo. Multi-level wavelet-cnn for image restoration. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 773–782, 2018.
46. Guanyu Li, Xiaoling Xu, Minghui Zhang, and Qiegen Liu. Densely connected network for impulse noise removal. *Pattern Analysis and Applications*, 23(3):1263–1275, 2020.
47. Chunwei Tian, Yong Xu, Zuoyong Li, Wangmeng Zuo, Lunke Fei, and Hong Liu. Attention-guided cnn for image denoising. *Neural Networks*, 124:117–129, 2020.
48. Ke Yu, Chao Dong, Chen Change Loy, and Xiaoou Tang. Deep convolution networks for compression artifacts reduction. *arXiv preprint arXiv:1608.02778*, 2016.
49. Yunjin Chen and Thomas Pock. Trainable nonlinear reaction diffusion: A flexible framework for fast and effective image restoration. *IEEE transactions on pattern analysis and machine intelligence*, 39(6):1256–1272, 2016.
50. Yongbing Zhang, Lulu Sun, Chenggang Yan, Xiangyang Ji, and Qionghai Dai. Adaptive residual networks for high-quality image restoration. *IEEE Transactions on Image Processing*, 27(7):3150–3163, 2018.
51. Hu Chen, Yi Zhang, Mannudeep K Kalra, Feng Lin, Yang Chen, Peixi Liao, Jiliu Zhou, and Ge Wang. Low-dose ct with a residual encoder-decoder convolutional neural network. *IEEE transactions on medical imaging*, 36(12):2524–2535, 2017.
52. Miao Yang and Arcot Sowmya. An underwater color image quality evaluation metric. *IEEE Transactions on Image Processing*, 24(12):6062–6071, 2015.
53. Karen Panetta, Chen Gao, and Sos Agaian. Human-visual-system-inspired underwater image quality measures. *IEEE Journal of Oceanic Engineering*, 41(3):541–551, 2015.
54. Eirikur Agustsson and Radu Timofte. Ntire 2017 challenge on single image super-resolution: Dataset and study. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 126–135, 2017.
55. Pablo Arbelaez, Michael Maire, Charless Fowlkes, and Jitendra Malik. Contour detection and hierarchical image segmentation. *IEEE transactions on pattern analysis and machine intelligence*, 33(5):898–916, 2010.

56. Woong Bae, Jaejun Yoo, and Jong Chul Ye. Beyond deep residual learning for image restoration: Persistent homology-guided manifold simplification. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 145–153, 2017.
57. Xin Yu, Peng Dai, Wenbo Li, Lan Ma, Jiajun Shen, Jia Li, and Xiaojuan Qi. Towards efficient and scale-robust ultra-high-definition image demoiréing. In *European Conference on Computer Vision*, pages 646–662. Springer, 2022.
58. Zhendong Wang, Xiaodong Cun, Jianmin Bao, Wengang Zhou, Jianzhuang Liu, and Houqiang Li. Uformer: A general u-shaped transformer for image restoration. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 17683–17693, 2022.
59. Bin He, Ce Wang, Boxin Shi, and Ling-Yu Duan. Fhde 2 net: Full high definition demoireing network. In *European Conference on Computer Vision*, pages 713–729. Springer, 2020.
60. Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 136–144, 2017.
61. Xinjie Li, Guojia Hou, Kunqian Li, and Zhenkuan Pan. Enhancing underwater image via adaptive color and contrast enhancement, and denoising. *Engineering Applications of Artificial Intelligence*, 111:104759, 2022.
62. Zhiying Jiang, Zhuoxiao Li, Shuzhou Yang, Xin Fan, and Risheng Liu. Target oriented perceptual adversarial fusion network for underwater image enhancement. *IEEE Transactions on Circuits and Systems for Video Technology*, 32(10):6584–6598, 2022.
63. Chongyi Li, Saeed Anwar, and Fatih Porikli. Underwater scene prior inspired deep underwater image and video enhancement. *Pattern Recognition*, 98:107038, 2020.
64. Xiaohong Yan, Guangxin Wang, Guangyuan Wang, Yafei Wang, and Xianping Fu. A novel biologically-inspired method for underwater image enhancement. *Signal Processing: Image Communication*, 104:116670, 2022.
65. Alejandra Sierra Jussi Tohka Morteza Ghahremani, Mohammad Khateri. Adversarial distortion learning for medical image denoising. *arXiv:2204.14100*, 2022.
66. Jingwen He, Chao Dong, and Yu Qiao. Modulating image restoration with continual levels via adaptive feature modification layers. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 11056–11064, 2019.
67. Hanting Chen, Yunhe Wang, Tianyu Guo, Chang Xu, Yiping Deng, Zhenhua Liu, Siwei Ma, Chunjing Xu, Chao Xu, and Wen Gao. Pre-trained image processing transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12299–12310, 2021.
68. Gregory Vaksman, Michael Elad, and Peyman Milanfar. Patch craft: Video denoising by deep modeling and patch matching. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2157–2166, 2021.
69. Gregory Vaksman, Michael Elad, and Peyman Milanfar. Lidia: Lightweight learned image denoising with instance adaptation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 524–525, 2020.
70. Majed El Helou and Sabine Süsstrunk. Blind universal bayesian image denoising with gaussian noise level learning. *IEEE Transactions on Image Processing*, 29:4885–4897, 2020.
71. Dror Simon and Michael Elad. Rethinking the csc model for natural images. *Advances in Neural Information Processing Systems*, 32:2271–2281, 2019.
72. Saeed Anwar and Nick Barnes. Real image denoising with feature attention. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 3155–3164, 2019.
73. Michele Claus and Jan van Gemert. Videnn: Deep blind video denoising. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pages 0–0, 2019.
74. Xianming Liu, Xiaolin Wu, Jiantao Zhou, and Debin Zhao. Data-driven soft decoding of compressed images in dual transform-pixel domain. *IEEE Transactions on Image Processing*, 25(4):1649–1659, 2016.
75. Chen Zhao, Jian Zhang, Siwei Ma, Xiaopeng Fan, Yongbing Zhang, and Wen Gao. Reducing image compression artifacts by structural sparse representation and quantization constraint prior. *IEEE Transactions on Circuits and Systems for Video Technology*, 27(10):2057–2071, 2016.
76. Honggang Chen, Xiaohai He, Linbo Qing, Shuhua Xiong, and Truong Q Nguyen. Dpw-sdnet: Dual pixel-wavelet domain deep cnns for soft decoding of jpeg-compressed images. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition Workshops*, pages 711–720, 2018.
77. Mengdi Sun, Xiaohai He, Shuhua Xiong, Chao Ren, and Xinglong Li. Reduction of jpeg compression artifacts based on dct coefficients prediction. *Neurocomputing*, 384:335–345, 2020.